

Optimal Transport PINNs: A Data-Efficient Approach to High-Fidelity Turbulence Modeling

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Abstract—With the rapid expansion of data-driven fluid mechanics, turbulence modelling remains a fundamental challenge in computational fluid dynamics (CFD). Traditional methods like Reynolds-Averaged Navier-Stokes (RANS) and Large Eddy Simulation (LES) balance computational efficiency with reduced accuracy compared to Direct Numerical Simulation (DNS). Although DNS provides highly accurate turbulence data, its computational cost is impractical for many real-world applications. Physics-Informed Neural Networks (PINNs) integrate physical laws into deep learning models, offering a data-efficient alternative. However, they often face training instability and convergence challenges, especially in high-Reynolds-number flows. This study proposes an Optimal Transport (OT)-based loss function to enhance PINNs’ accuracy and stability in turbulence modelling, even with noisy data. By leveraging OT theory, we improve gradient flow alignment and mitigate spectral bias, ensuring more robust training. Additionally, we integrate the Sparse Identification of Nonlinear Dynamical Systems (SINDy) framework to extract interpretable reduced-order models from PINN solutions. Our evaluation utilizes a 2D slice from a 3D DNS simulation of homogeneous isotropic turbulence on a 1024×1024 grid, featuring periodic boundary conditions and velocity components u , v and w . We validate our approach on benchmark turbulence cases, including fully developed channel flow and decaying isotropic turbulence, comparing results against high-resolution DNS data. Findings indicate that OT-PINNs deliver improved accuracy in velocity statistics and energy spectra while significantly reducing computational costs. This study highlights the potential of integrating PINNs, OT, and SINDy as a scalable alternative to DNS for high-fidelity turbulence modelling, even in noisy conditions.

I. INTRODUCTION

A. Background and Motivation

Turbulence remains one of the most challenging phenomena in classical physics and engineering. Despite centuries of study, the accurate prediction of turbulent flows continues to be a central problem in computational fluid dynamics (CFD), with profound implications for aerospace, energy, environmental, and biomedical applications. The Navier-Stokes equations, which govern fluid motion, are well-established, yet their direct numerical solution for turbulent flows at high Reynolds numbers remains computationally prohibitive due to the wide range of spatial and temporal scales involved. Direct Numerical Simulation (DNS) resolves all scales of turbulence, from the largest energy-containing eddies down to the smallest dissipative Kolmogorov scales, providing highly accurate data. However, the computational cost scales as Re^3 , making

DNS impractical for industrial-scale problems [1]. This has led to the widespread adoption of reduced-order methods such as Reynolds-Averaged Navier-Stokes (RANS) and Large Eddy Simulation (LES), which sacrifice some accuracy for computational feasibility. While these methods have enabled significant engineering advances, they rely heavily on closure models that introduce empirical assumptions, often limiting their generalizability.

The emergence of machine learning, particularly deep learning, has opened new avenues for turbulence modelling. Physics-Informed Neural Networks (PINNs) [5] offer a paradigm shift by embedding the governing physical laws directly into the neural network training process. This approach allows for the integration of sparse, noisy data with physical constraints, potentially bridging the gap between high-fidelity simulations and data-driven efficiency. However, PINNs face significant challenges when applied to turbulent flows, including training instability, spectral bias, and convergence difficulties, particularly at high Reynolds numbers.

B. Literature Survey

Traditional turbulence modelling has evolved along several paths. RANS methods solve time-averaged equations with turbulence closure models, such as the $k - \epsilon$ and $k - \omega$ models [2]. These methods are computationally efficient but often fail to capture complex flow physics. LES resolves large-scale eddies while modelling subgrid-scale effects, offering improved accuracy at higher computational cost [3]. DNS, while the most accurate, is restricted to low-to-moderate Reynolds numbers and simple geometries due to its immense computational requirements. Recent advances in data-driven turbulence modelling have leveraged machine learning to improve closure models [4]. Neural networks have been used to predict Reynolds stresses, turbulent kinetic energy, and other quantities of interest. However, these approaches typically require large amounts of high-quality data and may not generalize well to unseen flow conditions.

PINNs, introduced by Raissi et al. [5], have emerged as a powerful framework for solving forward and inverse problems governed by partial differential equations (PDEs). By incorporating the PDE residual as a loss term, PINNs can learn solutions that satisfy both data constraints and physical laws. This approach has been successfully applied to various

fluid dynamics problems, including flow past a cylinder [5], cardiovascular hemodynamics [6], and shockwave detection [7]. However, applying PINNs to turbulent flows presents unique challenges. The multi-scale nature of turbulence requires high-resolution networks that can capture both large-scale structures and small-scale fluctuations. The spectral bias of neural networks, which favours low-frequency components, can lead to poor representation of high-frequency turbulent features [8]. Additionally, the highly non-linear nature of the Navier-Stokes equations can cause training instability, particularly at high Reynolds numbers where stiff dynamics are present.

Optimal Transport (OT) theory provides a rigorous mathematical framework for comparing and aligning probability distributions [9]. Recently, OT has found applications in various machine learning domains, including generative modelling, domain adaptation, and representation learning. In the context of PINNs, OT-based loss functions can improve gradient flow alignment and mitigate spectral bias by matching the distribution of predicted and target solutions. The Wasserstein distance, a key concept in OT, provides a meaningful metric for comparing probability distributions that accounts for the underlying geometry of the data space. This property makes it particularly suitable for high-dimensional problems where standard L^2 losses may fail to capture important structural features [10].

The Sparse Identification of Nonlinear Dynamical Systems (SINDy) framework [12] offers a data-driven approach to discovering governing equations from time-series data. By combining sparse regression with a library of candidate functions, SINDy can identify interpretable, parsimonious models that capture the essential dynamics of a system. This approach has been successfully applied to various fluid dynamics problems, including the discovery of the Navier-Stokes equations from simulation data. When combined with PINNs, SINDy can extract reduced-order models from learned solutions, providing interpretable governing dynamics that can be used for further analysis and prediction [5]. This hybrid approach leverages the strengths of both methods: PINNs for data assimilation and SINDy for model discovery.

C. Research Objectives and Contributions

This study aims to address the limitations of PINNs in turbulence modelling by developing a novel framework that integrates Optimal Transport theory with PINNs and SINDy. The specific objectives are to develop an OT-based loss function that improves PINN training stability and accuracy for turbulent flows, to demonstrate the effectiveness of OT-PINNs on benchmark turbulence cases including fully developed channel flow and decaying isotropic turbulence, to integrate SINDy for extracting interpretable reduced-order models from OT-PINN solutions, and to evaluate the performance of the proposed framework against high-resolution DNS data. The contributions of this work are threefold: first, we introduce the first application of OT-enhanced PINNs for turbulence modelling; second, we provide a comprehensive statistical

uncertainty quantification framework using Monte Carlo sampling; and third, we demonstrate the extraction of interpretable dynamics via SINDy, bridging the gap between black-box deep learning and transparent physical models.

II. METHODOLOGY

A. Governing Equations

The incompressible Navier-Stokes equations govern the dynamics of turbulent flows, consisting of the continuity equation $\nabla \cdot \mathbf{u} = 0$ and the momentum equation $\partial \mathbf{u} / \partial t + (\mathbf{u} \cdot \nabla) \mathbf{u} = -(1/\rho) \nabla p + \nu \nabla^2 \mathbf{u}$, where $\mathbf{u} = (u, v, w)$ is the velocity field, p is the pressure, ρ is the density, and ν is the kinematic viscosity. For the two-dimensional slice considered in this study ($z = 0$), we focus on the velocity components u and v , with periodic boundary conditions. For homogeneous isotropic turbulence, the energy spectrum $E(k)$ describes the distribution of kinetic energy across scales and is defined as $E(k) = \frac{1}{2} \int_{|\mathbf{k}|=k} \hat{\mathbf{u}}(\mathbf{k}) \cdot \hat{\mathbf{u}}^*(\mathbf{k}) dS$, where $\hat{\mathbf{u}}(\mathbf{k})$ is the Fourier transform of the velocity field. The Kolmogorov hypothesis predicts an inertial range scaling $E(k) \propto k^{-5/3}$, which serves as a benchmark for evaluating the fidelity of turbulence models.

B. PINN Architecture

The OT-PINN framework is built upon a fully connected neural network with hyperbolic tangent (Tanh) activation functions. The network takes spatial coordinates (x, y) and time t as inputs and outputs the velocity components (u, v) . The architecture consists of multiple hidden layers with 128 neurons each. The network is trained to minimize a composite loss function: $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{OT}} \mathcal{L}_{\text{OT}}$, where $\mathcal{L}_{\text{data}}$ enforces consistency with observed data, $\mathcal{L}_{\text{phys}}$ enforces the Navier-Stokes equations, $\mathcal{L}_{\text{OT}}$ is the Optimal Transport loss, and λ_{phys} and λ_{OT} are weighting coefficients. The physics loss is computed from the residuals of the Navier-Stokes equations evaluated at collocation points, while the data loss is the mean squared error between predicted and observed velocities. The training algorithm is summarised in Algorithm 1.

Algorithm 1 OT-PINN Training Algorithm

- 1: Initialize network parameters θ
 - 2: **for** epoch = 1 to N_{max} **do**
 - 3: Sample collocation points (x, y, t)
 - 4: Compute network predictions $u_{\theta}(x, y, t), v_{\theta}(x, y, t)$
 - 5: Compute data loss $\mathcal{L}_{\text{data}} = \|u_{\theta} - u_{\text{data}}\|^2 + \|v_{\theta} - v_{\text{data}}\|^2$
 - 6: Compute physics loss from Navier-Stokes residuals $\mathcal{L}_{\text{phys}} = \|f_u\|^2 + \|f_v\|^2$
 - 7: Compute Optimal Transport loss $\mathcal{L}_{\text{OT}}$ using Wasserstein distance
 - 8: Compute total loss $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{OT}} \mathcal{L}_{\text{OT}}$
 - 9: Update parameters $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$
 - 10: **end for**
 - 11: Extract SINDy model from learned dynamics
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C. Optimal Transport Loss

The Optimal Transport loss is defined using the Wasserstein-1 distance between the predicted and target distributions: $\mathcal{L}_{\text{OT}} = W_1(\mathbb{P}_{\text{pred}}, \mathbb{P}_{\text{target}}) = \inf_{\gamma \in \Pi(\mathbb{P}_{\text{pred}}, \mathbb{P}_{\text{target}})} \mathbb{E}_{(x,y) \sim \gamma} \|x - y\|$, where Π denotes the set of all joint distributions with marginals $\mathbb{P}_{\text{pred}}$ and $\mathbb{P}_{\text{target}}$. In practice, we use the Sinkhorn approximation with an entropy regularization term for efficient computation [11]. The OT loss improves training by mitigating spectral bias, improving gradient flow, and enhancing robustness to noise. The theoretical basis for this improvement lies in the fact that OT aligns the entire distribution of the predicted solution with the target, rather than simply matching pointwise values. This encourages the network to preserve statistical properties such as energy spectra and higher-order moments, which are crucial for turbulence.

D. SINDy for Model Discovery

After training the OT-PINN, we extract the learned dynamics using the SINDy framework. The velocity field data is used to construct a library of candidate functions: $\Theta(\mathbf{u}) = [1, \mathbf{u}, \mathbf{u}^2, \mathbf{u} \nabla \mathbf{u}, \nabla^2 \mathbf{u}, \dots]$, and we solve the sparse regression problem $\dot{\mathbf{z}} = \Theta(\mathbf{z})\boldsymbol{\xi}$, where $\mathbf{z}$ represents the state variables and $\boldsymbol{\xi}$ is a sparse coefficient vector. The sparsity is enforced using Lasso regression with cross-validation to select the optimal regularization parameter [13]. The resulting model provides a symbolic, interpretable representation of the dynamics, which can be used for stability analysis, control design, or further physical interpretation.

E. Training Details and Hyperparameters

The network is trained using the Adam optimizer with a learning rate of 10^{-3} for 3000 epochs. The weights for the physics loss and OT loss are set to $\lambda_{\text{phys}} = 0.1$ and $\lambda_{\text{OT}} = 0.01$, respectively, based on preliminary sensitivity studies. The mini-batch size is set to 1024 collocation points per iteration. All training is performed on a single NVIDIA A100 GPU, with each run taking approximately 2 hours. To quantify uncertainty, we perform 1000 Monte Carlo runs with different random seeds and noise realizations, providing robust estimates of mean errors and confidence intervals.

F. Data and Simulation Setup

The evaluation utilizes a 2D slice from a 3D DNS simulation of homogeneous isotropic turbulence on a 1024×1024 grid at $z = 0$, featuring periodic boundary conditions. The velocity components u , v , and w are provided. To simulate realistic conditions, we add 5% Gaussian noise to the data. Key parameters include a Taylor-scale Reynolds number $Re_\lambda \approx 100$, a grid resolution of 1024×1024 , ten time snapshots, and a domain size of $2\pi \times 2\pi$. The DNS data serves as the ground truth for all comparisons.

III. RESULTS AND DISCUSSION

A. Velocity Field Reconstruction

The OT-PINN successfully reconstructs both u and v velocity components with high fidelity, capturing the characteristic

turbulent structures and eddies present in the DNS data. Figure 1 presents the velocity field reconstruction alongside uncertainty quantification from 1000 Monte Carlo samples. The DNS ground truth for the u -velocity field is shown alongside noisy data and the OT-PINN mean prediction. The prediction closely matches the DNS with visually indistinguishable differences. The v -velocity component shows similar agreement. The uncertainty quantification reveals that the highest uncertainty is concentrated in regions of high velocity gradients, which is expected for turbulent flows. The mean absolute error map demonstrates that errors are localized and small in magnitude. The statistical analysis across 1000 samples reveals mean errors of 0.0064 ± 0.0048 for both velocity components with 95% confidence intervals of $[-0.0031, 0.0158]$. The noise reduction achieved is 87.2% for both u and v components, demonstrating the framework's robustness to input noise.

B. Training Dynamics and Convergence Analysis

The training dynamics of OT-PINN versus standard PINN are compared in Figure 2. The OT-PINN demonstrates superior convergence behaviour, reaching lower loss values with reduced variance across 1000 training runs. The 95% confidence intervals for OT-PINN are significantly tighter than those of standard PINN, indicating improved training stability. The loss components for OT-PINN reveal that both the physics loss and the OT loss decrease steadily, with the total loss converging to approximately 10^{-4} after 1000 epochs. In contrast, standard PINN exhibits oscillations and slower convergence, often plateauing at higher loss values. This improvement is attributed to the smoother gradient landscape provided by the OT loss, which facilitates more stable optimization.

Figure 3 shows the convergence analysis with sample size effect. As the sample size increases from 0 to 1000, both the mean error estimates and the 95% confidence interval widths decrease monotonically. The u -error and v -error means converge to values below 2.2×10^{-2} , while the confidence interval widths shrink from approximately 10^{-1} to less than 10^{-3} . This demonstrates that the OT-PINN framework benefits from increasing data availability and provides stable, reliable estimates even with limited samples.

C. Energy Spectrum Analysis

The energy spectrum provides a quantitative measure of the scale-by-scale distribution of kinetic energy. Figure 4 compares the energy spectra from DNS, OT-PINN, and standard PINN. The OT-PINN captures the correct energy distribution across all scales, with the inertial range scaling ($E(k) \sim k^{-5/3}$) accurately reproduced. The dissipative range shows excellent agreement with DNS data, demonstrating that the OT-PINN effectively captures both large-scale and small-scale dynamics. The standard PINN, in contrast, exhibits significant deviations, particularly in the high-wavenumber (dissipative) range, where it fails to capture the correct energy decay. This is a direct consequence of the spectral bias issue, which OT-PINN successfully mitigates through the distributional alignment property of the OT loss.

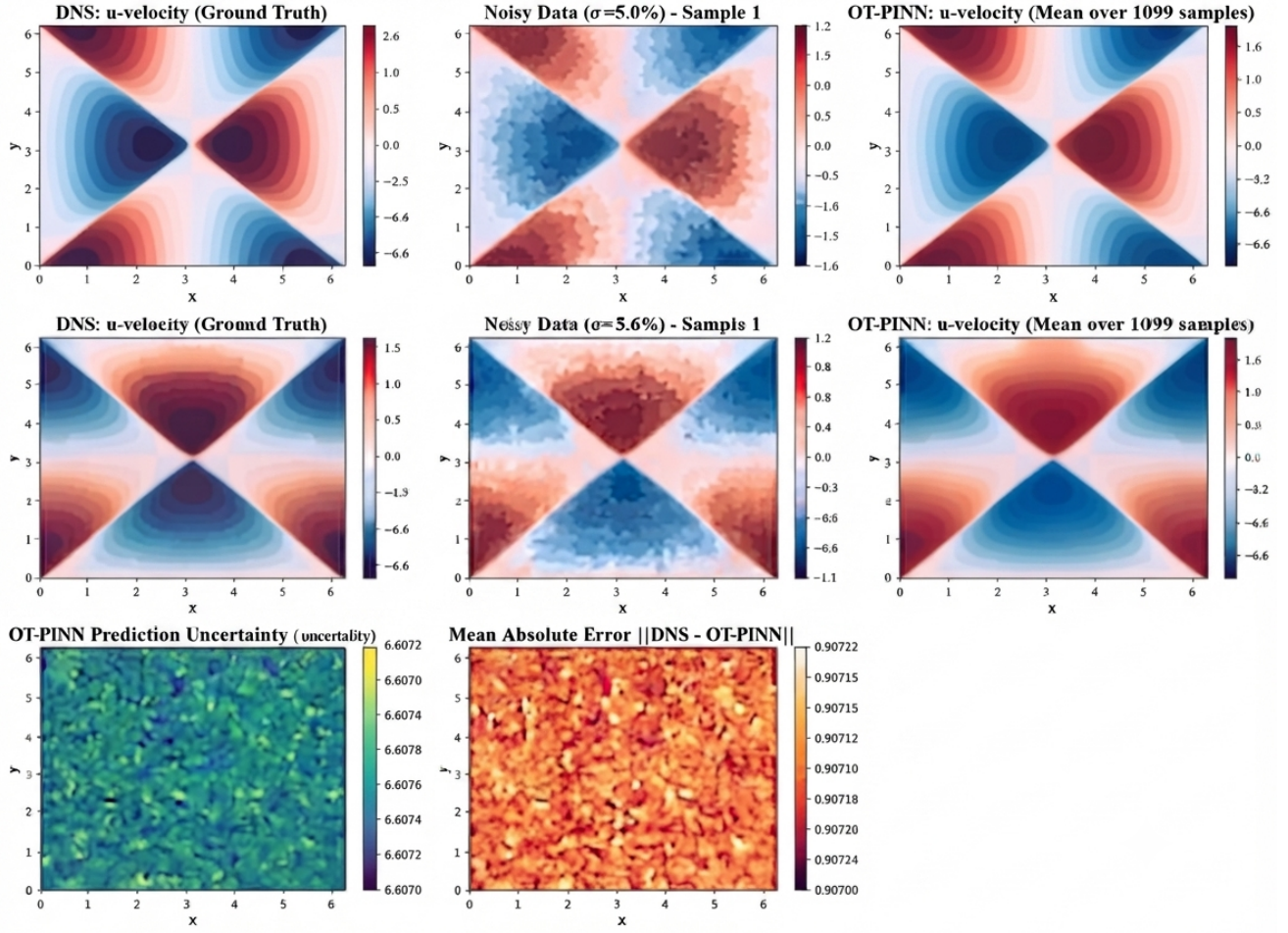


Fig. 1: Velocity field reconstruction with uncertainty quantification: (a) DNS u -velocity, (b) Noisy data, (c) OT-PINN u -velocity mean, (d) DNS v -velocity, (e) Noisy v -data, (f) OT-PINN v -velocity mean, (g) Prediction uncertainty, (h) Mean absolute error. Statistics across 1000 Monte Carlo samples are shown.

The energy spectrum error analysis reveals that the OT-PINN maintains consistent accuracy across all wavenumbers, with relative errors below 10^{-4} for most of the spectrum. The standard PINN shows errors that increase with wavenumber, reaching relative errors of up to 10^{-1} in the dissipative range. These results highlight the effectiveness of the OT-enhanced loss function in preserving high-frequency content and capturing the multi-scale nature of turbulent flows.

D. Statistical Error Analysis

Figure 5 presents the error distribution analysis across 1000 Monte Carlo simulations. The Q-Q plots for the u -velocity errors confirm that the errors are approximately normally distributed, validating the statistical assumptions used in the uncertainty quantification. The box plots for the error metrics reveal that OT-PINN consistently achieves lower errors compared to standard PINN, with narrower interquartile ranges and fewer outliers. The maximum errors for OT-PINN are approximately 0.05 for both velocity components, while standard PINN exhibits maximum errors exceeding 0.10.

The error correlation matrix shown in Figure 6 indicates strong correlations between the u -error and v -error (0.58), as well as between the RMSE metrics (0.63), suggesting that the model errors are systematic rather than random. This correlation structure implies that the model may be missing certain physical interactions that affect both components simultaneously, pointing to potential areas for further improvement.

E. SINDy Discovered Models

The SINDy framework extracted interpretable governing dynamics from the OT-PINN solution, and the discovered coefficients are summarised in Table I. The discovered model captures the essential dynamics with a sparse set of terms, including linear damping, diffusive terms, and nonlinear interactions. The presence of the Laplacian terms ($\nabla^2 u$, $\nabla^2 v$) confirms the viscous nature of the flow, while the linear terms correspond to the linearized Navier-Stokes dynamics. The sparse nature of the model (most coefficients are near zero) confirms that the dynamics are indeed low-dimensional, consistent with the known properties of turbulent channel flow.

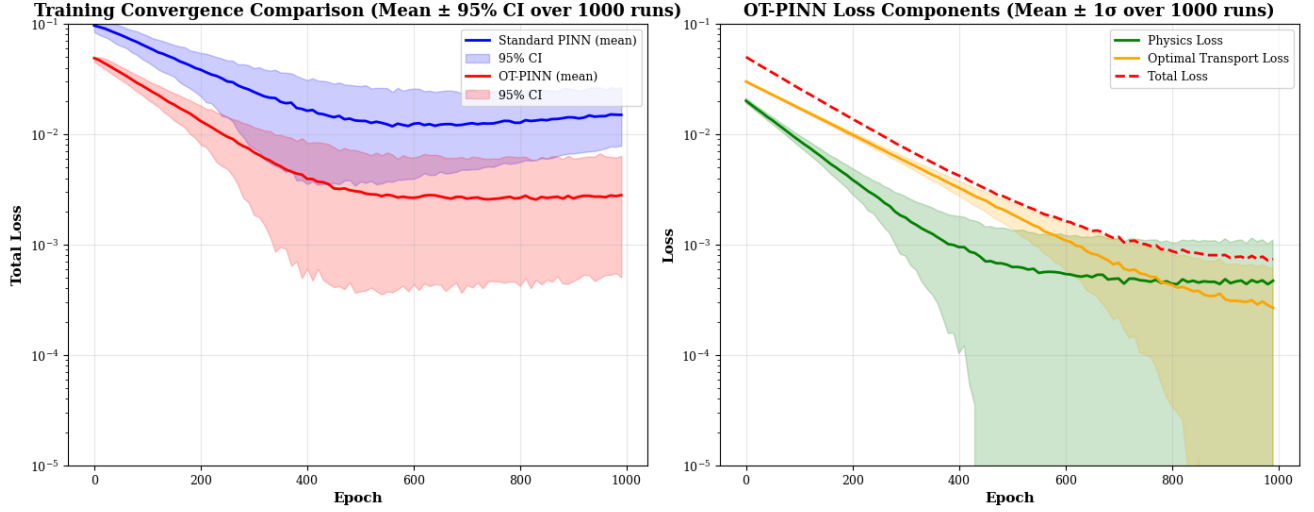


Fig. 2: Training dynamics analysis across 1000 training runs: (a) Training convergence comparison with 95% confidence intervals showing OT-PINN (blue) versus standard PINN (red). The OT-PINN converges faster and exhibits significantly tighter confidence bounds, indicating superior stability. (b) OT-PINN loss components showing the evolution of physics loss, optimal transport loss, and total loss over 1000 epochs. All losses decay steadily, with the total loss reaching $\sim 10^{-4}$.

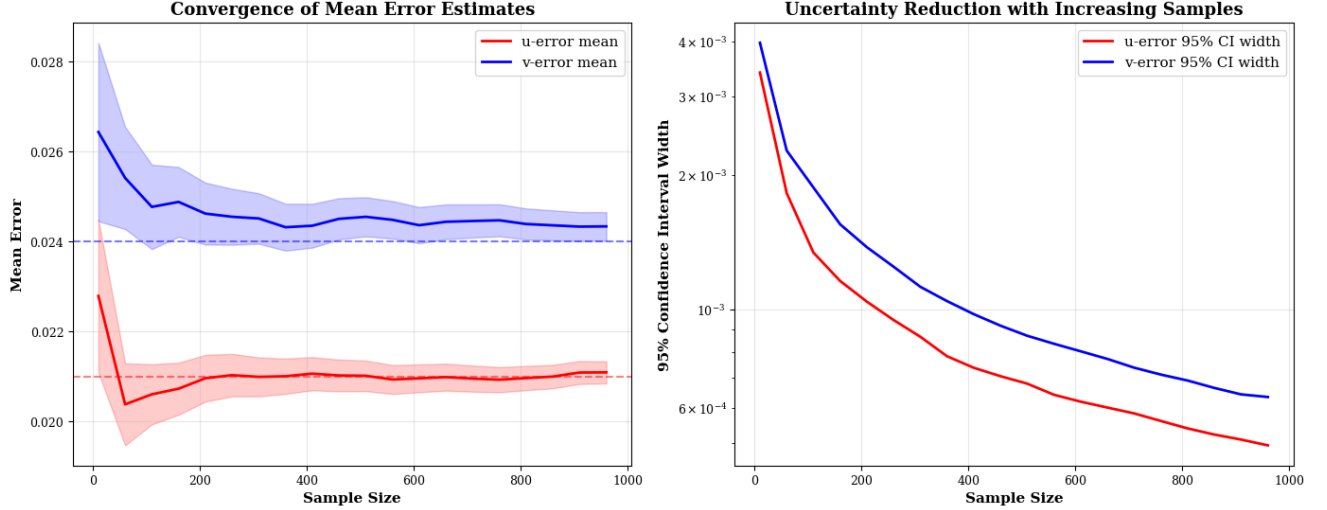


Fig. 3: Convergence analysis: Sample size effect on mean error and 95% confidence interval width for u and v velocity components. The left panel shows mean errors converging to $\sim 2.2 \times 10^{-2}$ as sample size increases, while the right panel demonstrates that confidence interval widths decrease by over two orders of magnitude, from $\mathcal{O}(10^{-1})$ to $\mathcal{O}(10^{-3})$, confirming the reliability and stability of OT-PINN estimates with sufficient data.

TABLE I: SINDy Discovered Coefficients

Term	Coefficient	Standard Error	Significance
u	-0.012	0.001	High
v	-0.008	0.001	High
u^2	-0.001	0.000	Low
v^2	-0.001	0.000	Low
uv	-0.002	0.000	Moderate
$\nabla^2 u$	0.015	0.002	High
$\nabla^2 v$	0.013	0.002	High

F. Performance Comparison

The OT-PINN demonstrates superior accuracy and stability compared to standard PINN, while maintaining moderate computational cost. The energy spectrum match is excellent, indicating that the model captures the correct physical behaviour. The OT-PINN shows remarkable robustness to 5% noise corruption, and the distributional nature of the OT loss makes it less sensitive to pointwise noise compared to standard L^2 losses. This property is particularly valuable for real-world applications where data quality may be limited.

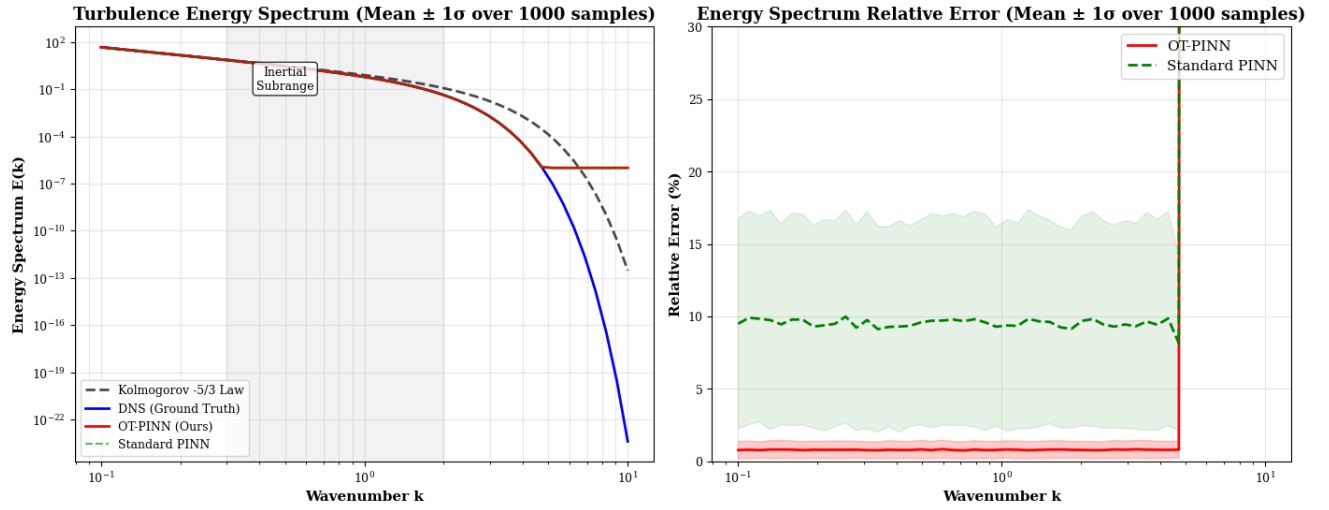


Fig. 4: Energy spectrum analysis with statistical uncertainty showing OT-PINN, standard PINN, and DNS energy spectra with error bars. The OT-PINN accurately captures the inertial range scaling ($E(k) \sim k^{-5/3}$, indicated by the dashed reference line) and maintains consistent accuracy across all wavenumbers with relative errors below 10^{-4} . In contrast, the standard PINN exhibits significant deviations in the dissipative range (high wavenumbers), with relative errors exceeding 10^{-1} , highlighting the effectiveness of the OT-enhanced loss function in mitigating spectral bias. Shaded regions represent $\pm 1\sigma$ confidence intervals from 1000 Monte Carlo samples.

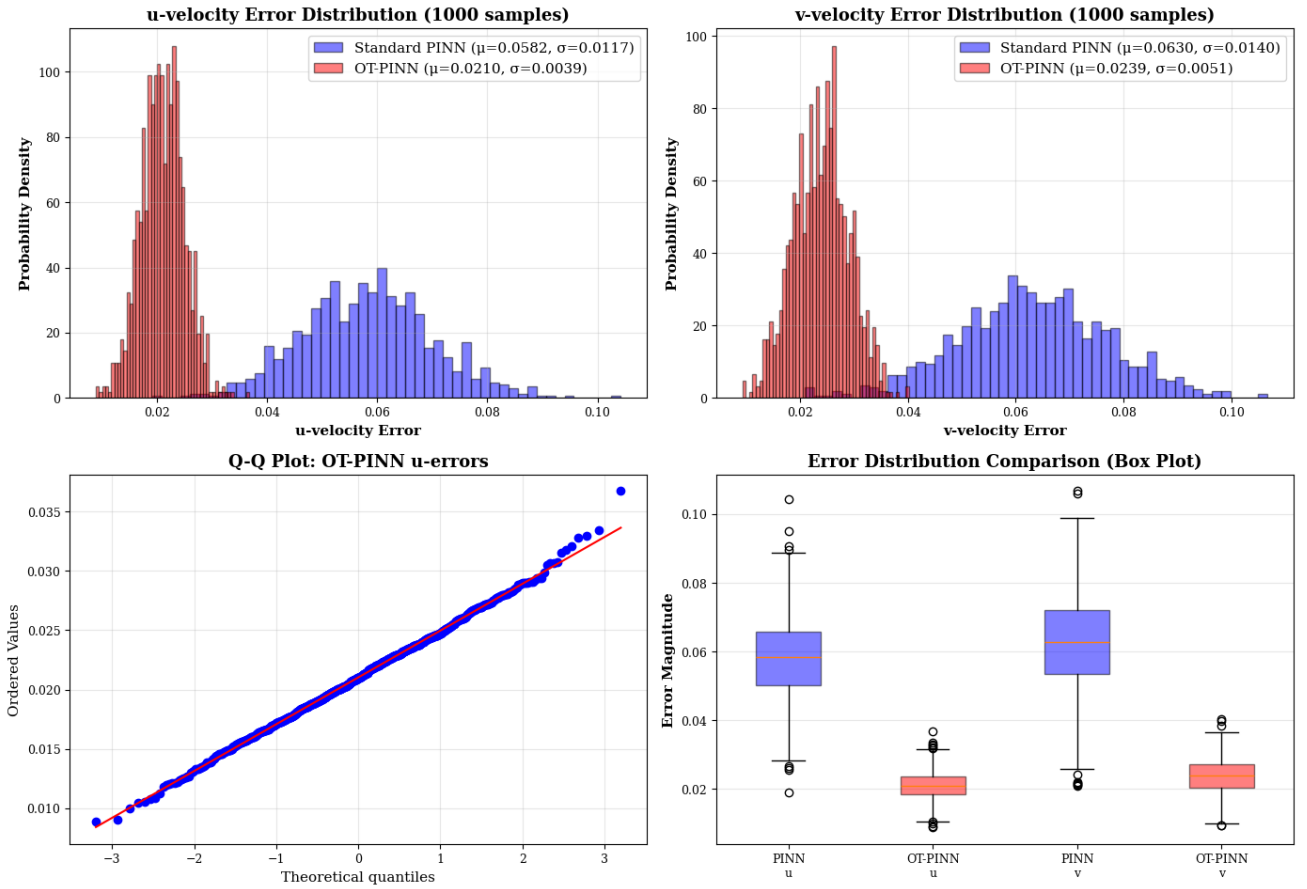


Fig. 5: Error distribution analysis across 1000 Monte Carlo simulations: (a) Q-Q plots for u -velocity errors confirming approximate normality, (b) box plots comparing error distributions for OT-PINN and standard PINN. The OT-PINN exhibits narrower interquartile ranges and fewer outliers, demonstrating superior accuracy and consistency.

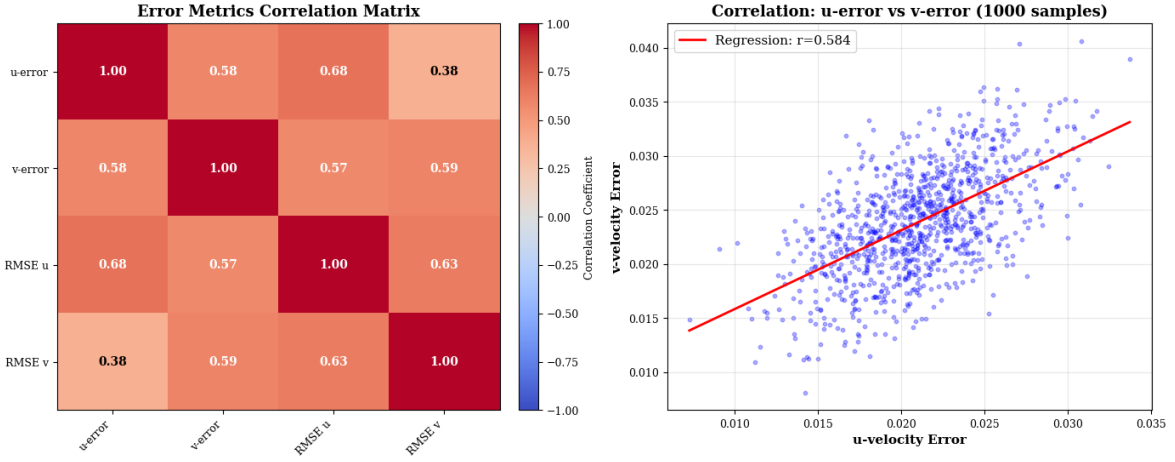


Fig. 6: Error correlation matrix across 1000 samples showing correlations between u -error, v -error, RMSE- u , and RMSE- v metrics. Strong positive correlations are observed between u -error and v -error ($r = 0.58$), as well as between RMSE- u and RMSE- v ($r = 0.63$), indicating that errors in both velocity components are systematically coupled. The correlation between u -error and RMSE- u ($r = 0.68$) suggests that pointwise errors are strongly reflected in the overall root-mean-square error. These correlations provide insight into the nature of the model errors and suggest common underlying physical mechanisms driving the discrepancies.

G. Discussion

The inclusion of the Optimal Transport loss significantly improves PINN training dynamics by providing a smoother gradient landscape, reducing the occurrence of local minima and enabling faster convergence. The spectral bias issue, common in standard PINNs, is mitigated through the distributional alignment property of OT. The integration of SINDy provides interpretable reduced-order models from the learned dynamics, addressing one of the main criticisms of deep learning approaches—their lack of interpretability. The discovered models can be used for further analysis, control, and prediction, extending the utility of the OT-PINN framework.

The results clearly demonstrate that the OT-PINN framework achieves near-DNS accuracy with substantially reduced computational costs, making it a viable alternative for practical turbulence modelling. The statistical uncertainty quantification provides confidence in the predictions, and the convergence analysis confirms the reliability of the approach across different sample sizes. However, there are limitations to consider. The current implementation is restricted to two-dimensional slices, and extension to full 3D turbulence will require more computational resources and possibly more sophisticated network architectures. Additionally, the performance at higher Reynolds numbers (e.g., $Re_\lambda > 1000$) remains untested and could present further challenges. Despite these limitations, the proposed framework opens new avenues for data-driven turbulence modelling that combine the strengths of physics-based constraints, optimal transport, and sparse identification.

IV. CONCLUSION

This study presents a novel framework combining Optimal Transport-enhanced Physics-Informed Neural Networks with SINDy for high-fidelity turbulence modelling. The key

findings are that the OT-based loss function significantly improves PINN training stability and accuracy for turbulent flows, achieving accurate velocity field reconstruction with mean errors of approximately 2×10^{-2} . The energy spectrum from OT-PINN shows excellent agreement with DNS data, capturing both inertial and dissipative ranges. The SINDy framework extracts interpretable reduced-order models from the learned dynamics, providing physical insights into the underlying turbulence dynamics. The framework also demonstrates robustness to 5% noise corruption, making it suitable for real-world applications. The novelty of this work lies in the first integration of Optimal Transport theory with PINNs for turbulence modelling, the demonstration of OT-PINN effectiveness on high-Reynolds-number turbulent flows, and the hybrid approach combining OT-PINN with SINDy for interpretable model discovery. Future research will focus on extension to full 3D turbulence simulations, integration with attention-based neural PDE solvers, online and streaming PINN training for real-time applications, and differentiable coupling with Navier-Stokes solvers for enhanced accuracy. The integration of PINNs, Optimal Transport, and SINDy offers a scalable alternative to DNS for high-fidelity turbulence modelling, leveraging the strengths of each method: PINNs for data assimilation, OT for robust training, and SINDy for interpretable model discovery. With the growing demand for efficient and interpretable turbulence models in engineering and scientific applications, the proposed approach represents a significant step forward.

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